MUCOPOLYSACCHARIDES

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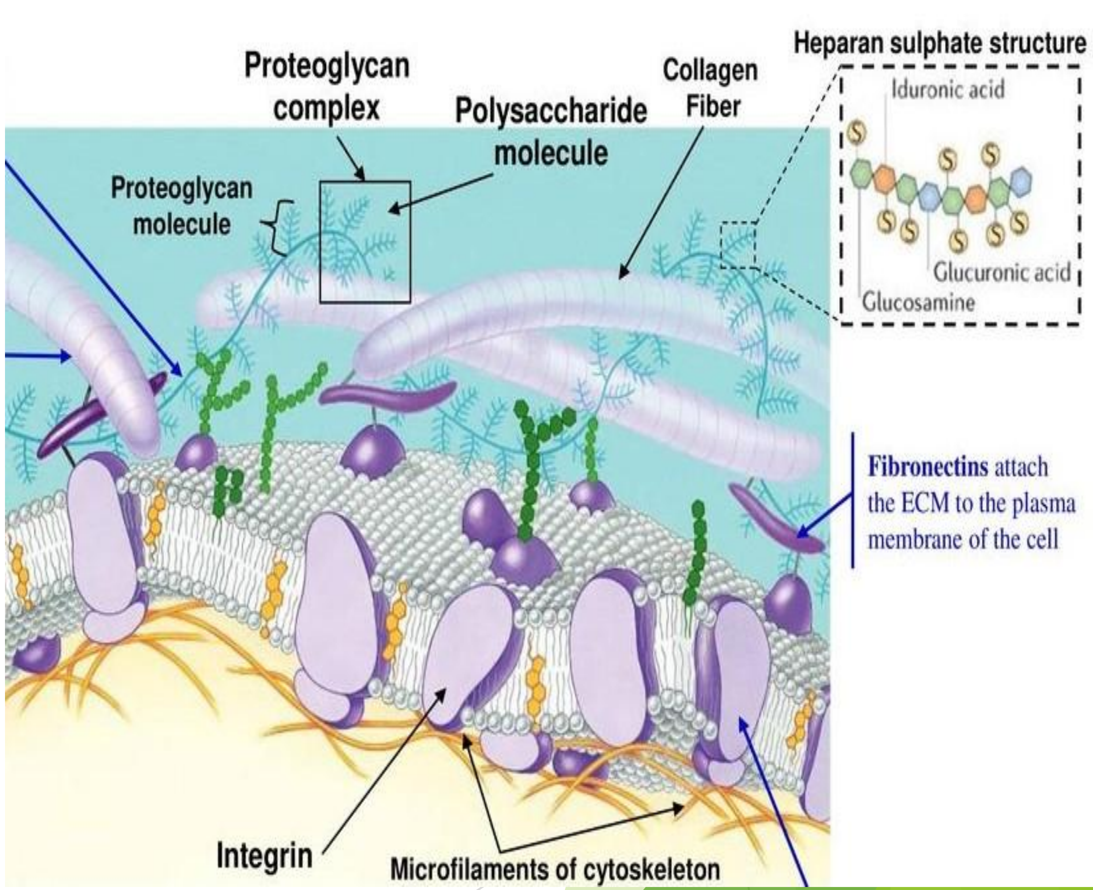
Introduction

- Mucopolysaccharides are a group of high molecular weight, linear polysaccharides.
- They are also known as Glycosaminoglycans (GAGs)
- They are major components of extracellular matrix (ECM), they have a number of important biologic roles and they are involved in a number of disease processes



- There are atleast seven GAGs

  (i) Hyaluronic acid (hyaluronan)

  (ii) Chondroitin sulfate

  (iii) Keratan sulfate I

  (iv) Keratan sulfate II

  (v) Heparin

  (vi) Heparan sulfate

  (vii) Dermatan sulfate

STRUCTURE OF MUCOPOLYSACCHARIDES

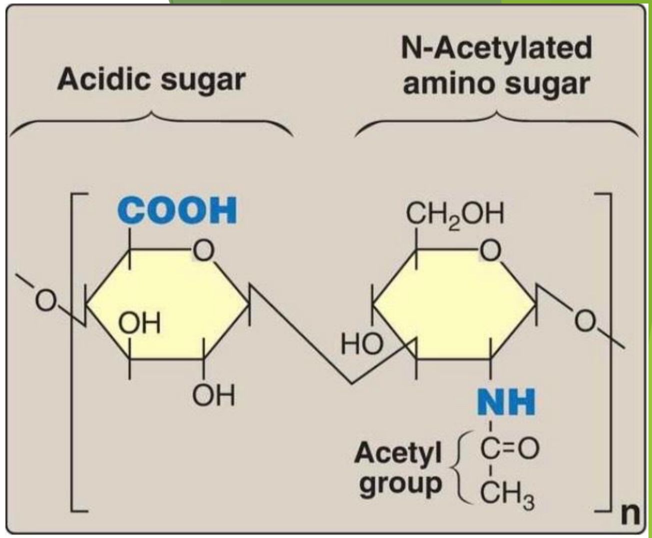
- The GAGs are made of repeating disaccharide units ; each unit contains –

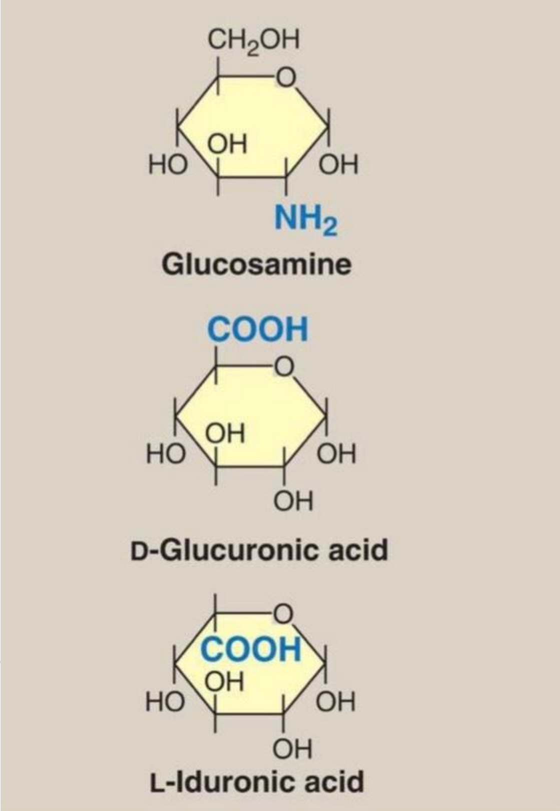
    (a) an amino sugar

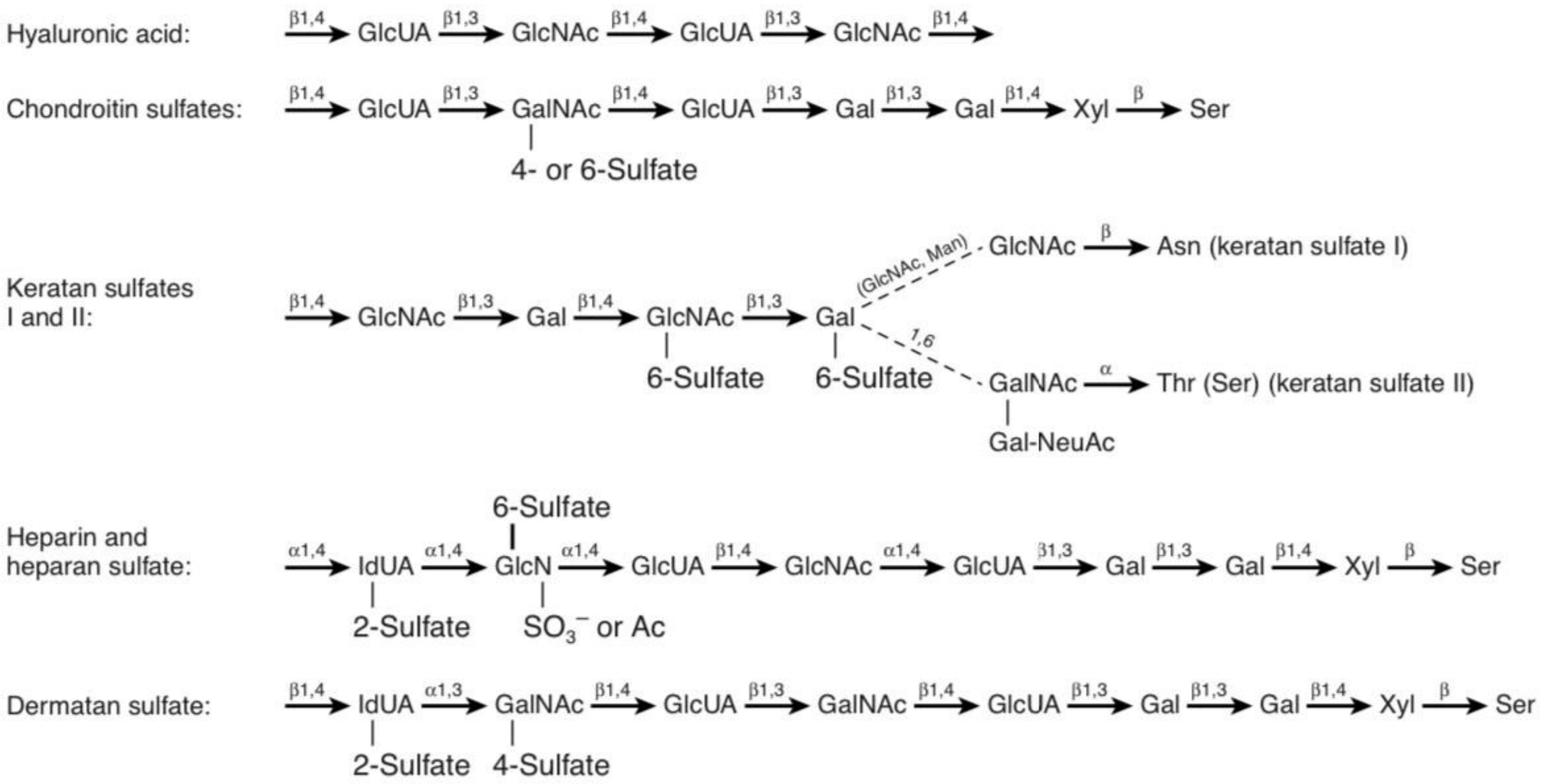
    (b) an acid sugar

- Amino sugar is either glucosamine or galactosamine

- Acid sugar is glucuronic in most cases and Iduronic acid in some cases



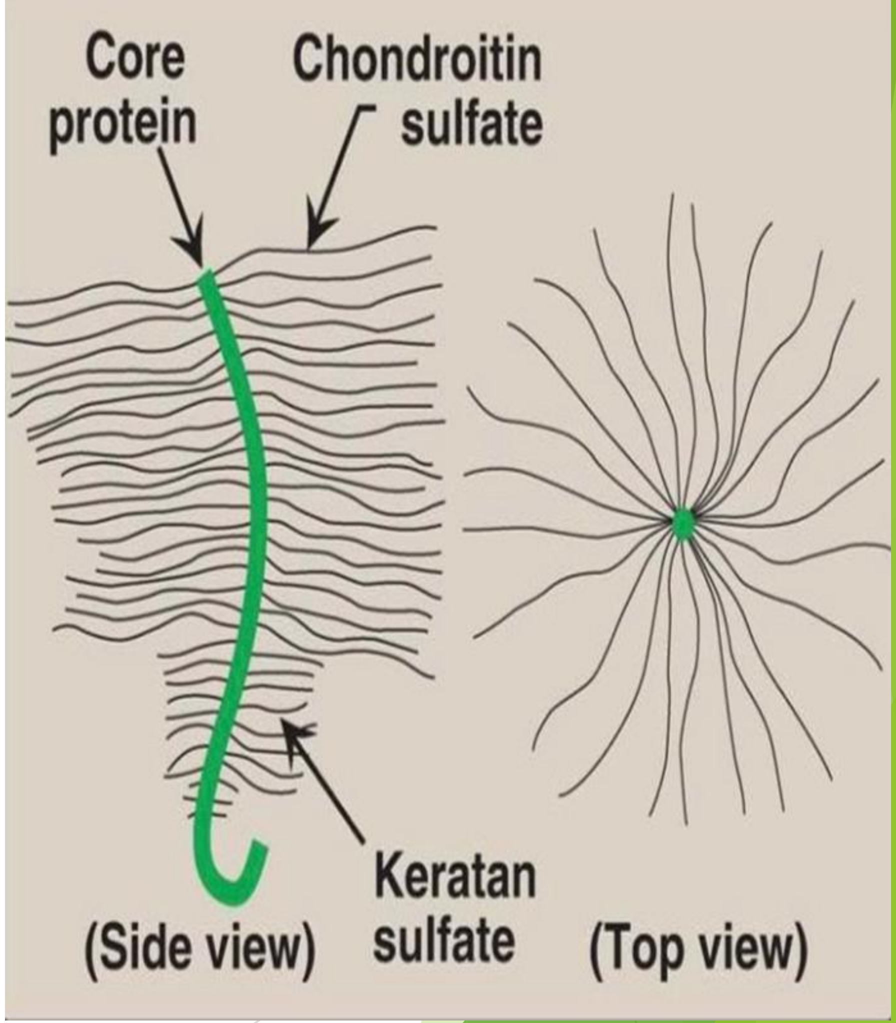


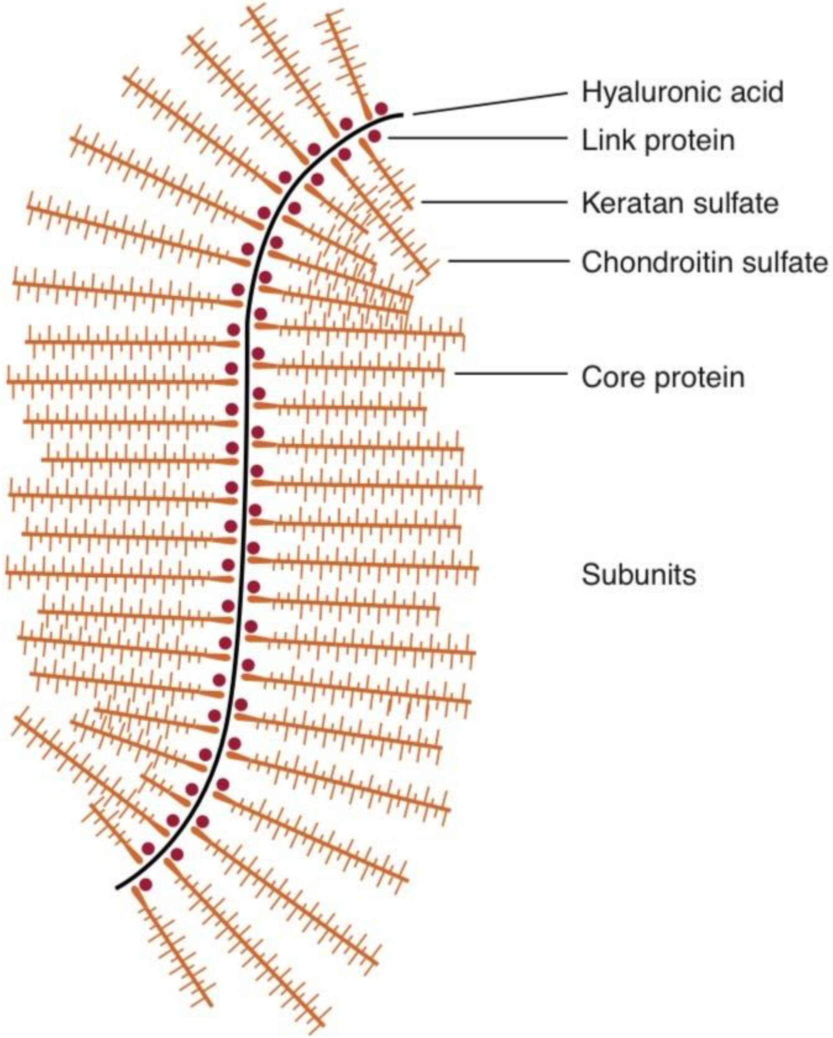

Hyaluronic acid:
β1,4 GlcUA β1,3 GlcNAc β1,4 GlcUA β1,3 GlcNAc β1,4
Chondroitin sulfates:
β1,4 GlcUA β1,3 GalNAc β1,4 GlcUA β1,3 Gal β1,3 Gal β1,4 Xyl β Ser
4- or 6-Sulfate
Keratan sulfates I and II:
β1,4 GlcNAc β1,3 Gal β1,4 GlcNAc β1,3 Gal
6-Sulfate
6-Sulfate
(GlcNAc, Man) GlcNAc β Asn (keratan sulfate I)
1,6 GalNAc α Thr (Ser) (keratan sulfate II)
Gal-NeuAc
Heparin and heparan sulfate:
α1,4 IdUA α1,4 GlcN α1,4 GlcUA β1,4 GlcNAc α1,4 GlcUA β1,3 Gal β1,3 Gal β1,4 Xyl β Ser
6-Sulfate
2-Sulfate
SO3− or Ac
Dermatan sulfate:
β1,4 IdUA α1,3 GalNAc β1,4 GlcUA β1,3 GalNAc β1,4 GlcUA β1,3 Gal β1,3 Gal β1,4 Xyl β Ser
2-Sulfate
4-Sulfate

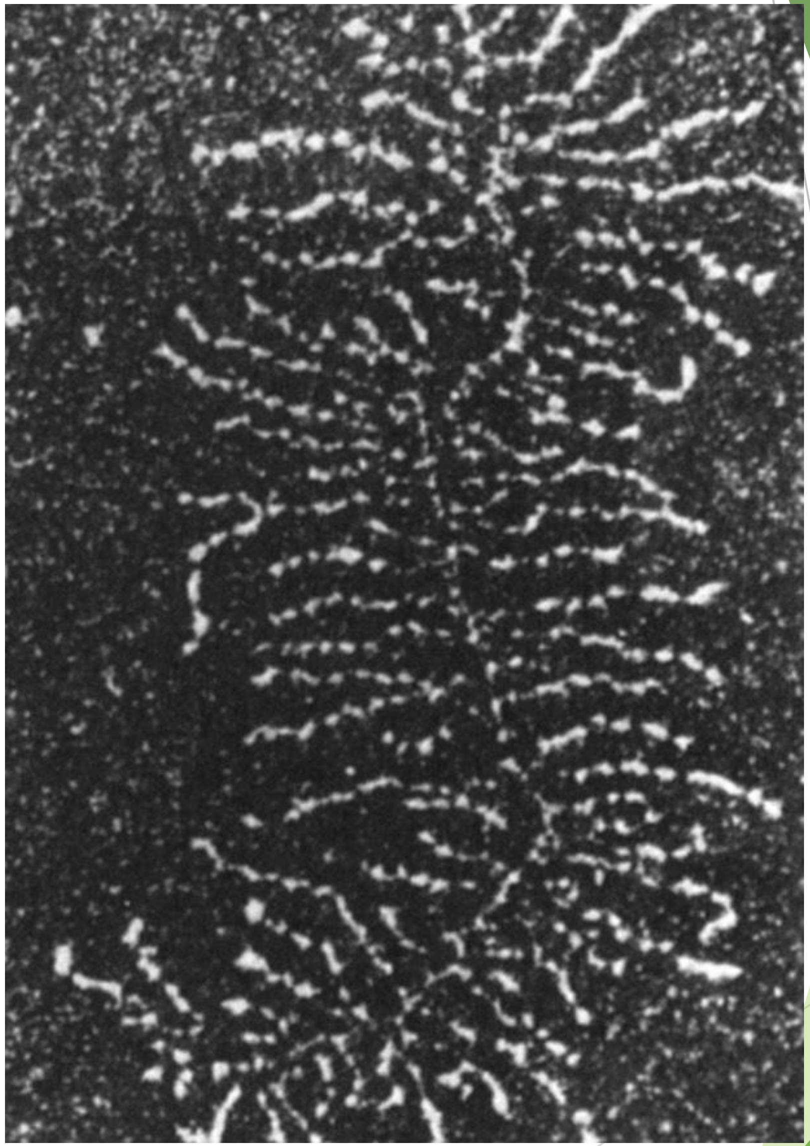
- They have complex structural organisations.

- GAGs are covalently attached to a protein (termed as core protein) to form hybrid molecules called proteoglycans, where the sugar constitutes upto 95% of proteoglycan molecule.

- The polysaccharide chains extend out from the core protein giving a ‘bottle brush’ appearance

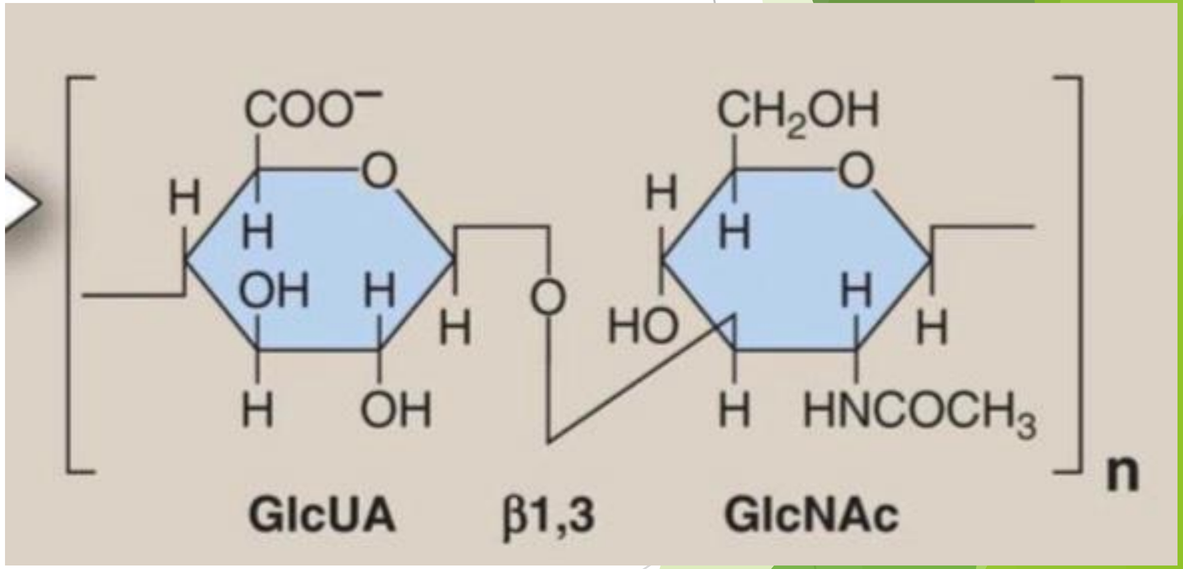



Hyaluronic acid
Link protein
Keratan sulfate
Chondroitin sulfate
Core protein
Subunits



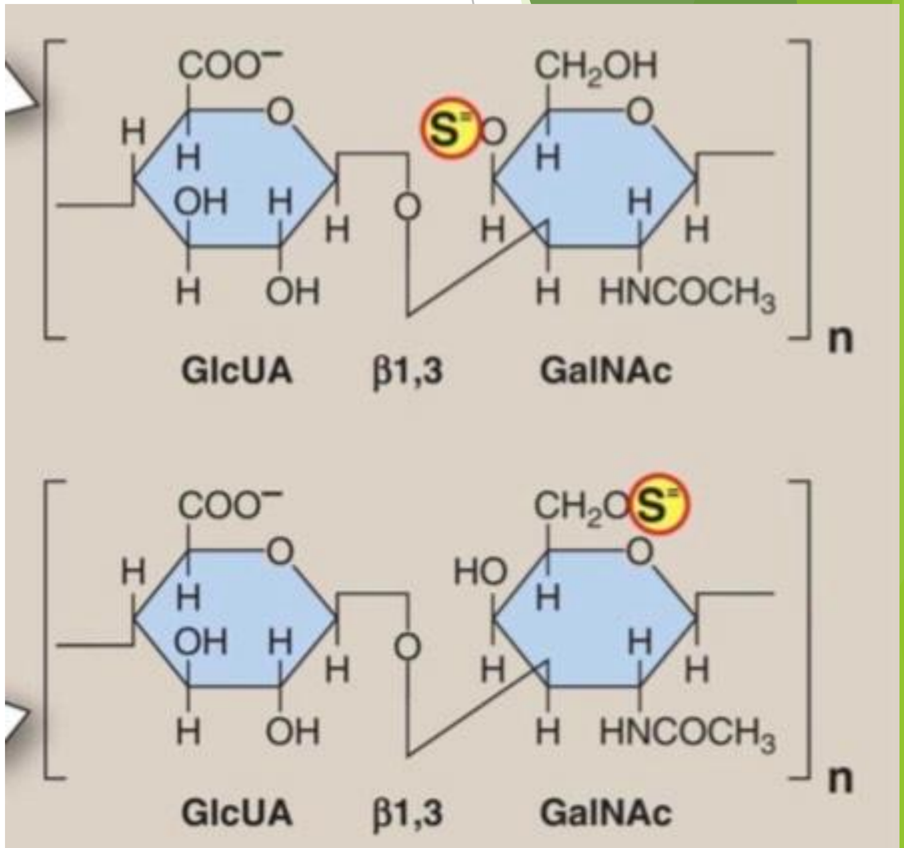
HYALURONIC ACID

- Contains repeating units of glucuronic acid and N-acetylglucosamine
- beta 1,3 and beta 1,4 glycosidic linkages
- Longest GAG
- No of disaccharides per chain ~50,000
- No sulfate groups and no protein linkages
- Present in synovial fluid of joints, cartilage, vitreous humour and connective tissues



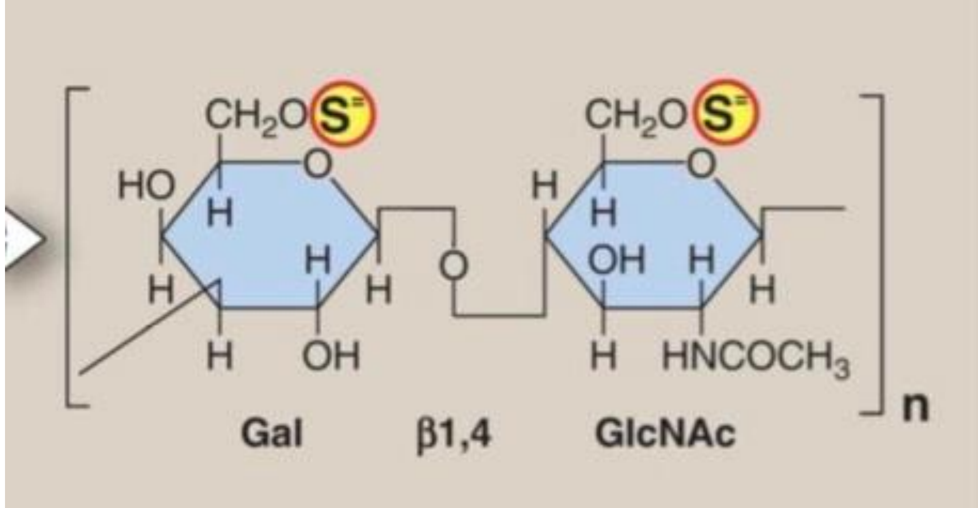
CHONDROITIN SULPHATE

- It is composed of repeating units of glucuronic acid and N-acetyl galactosamine sulphate
- No of disaccharides per chain = 20-60
- Beta 1,3 and beta 1,4 glycosidic linkages
- Present in ground substance of connective tissues
- Widely distributed in cartilage, tendons, bone and cornea.



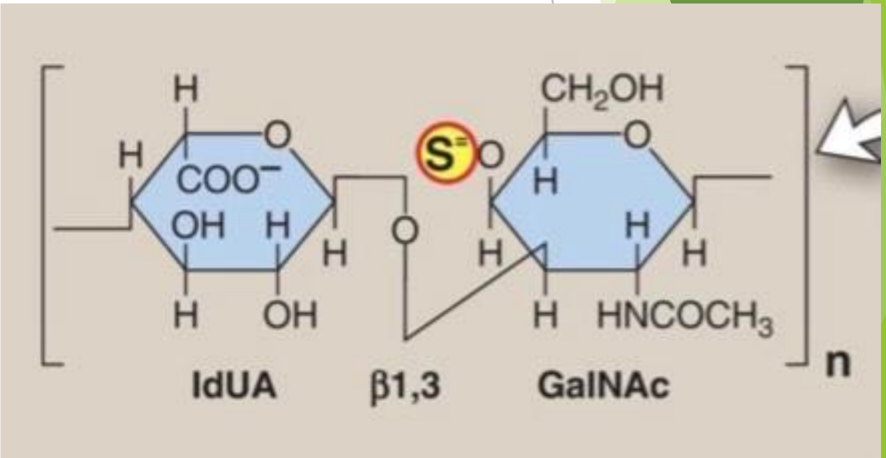
KERATAN SULPHATE

- It is an unusual GAG as it does not contain Acid sugar.
- The repeating disaccharide unit consists of N-acetylglucosamine and Galactose.
- No of disaccharides per chain ~25
- It is found in cornea and tendons.



DERMATAN SULPHATE

- The repeating disaccharide unit here is N-acetyl glucosamine and L-iduronic acid.
- It is found in skin, blood vessels and heart valves.



HEPARIN

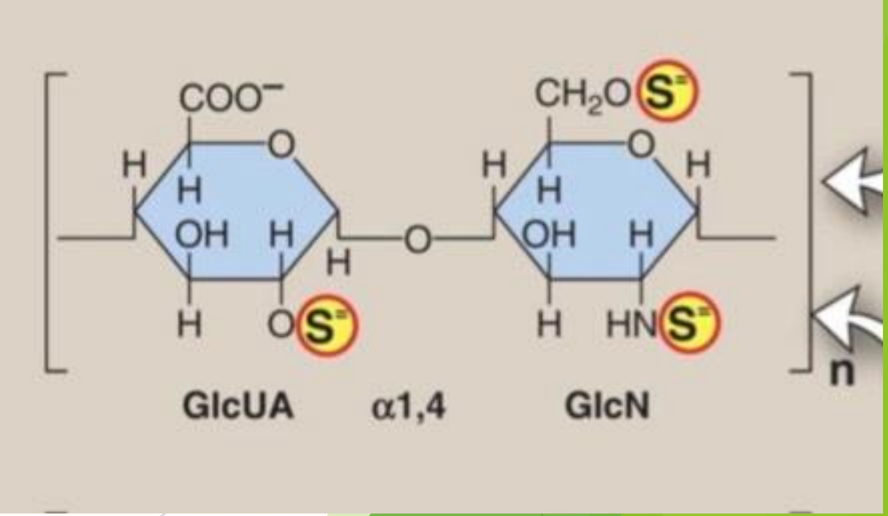
- The repeating disaccharide unit is the sulphated glucosamine and L-iduronic acid in alpha-1,4 linkages.

- No of disaccharides per chain = 15-90

- Heparin is an inhibitor of blood coagulation.

- Heparin is therapeutically used to prevent clotting in patients following a major surgery.

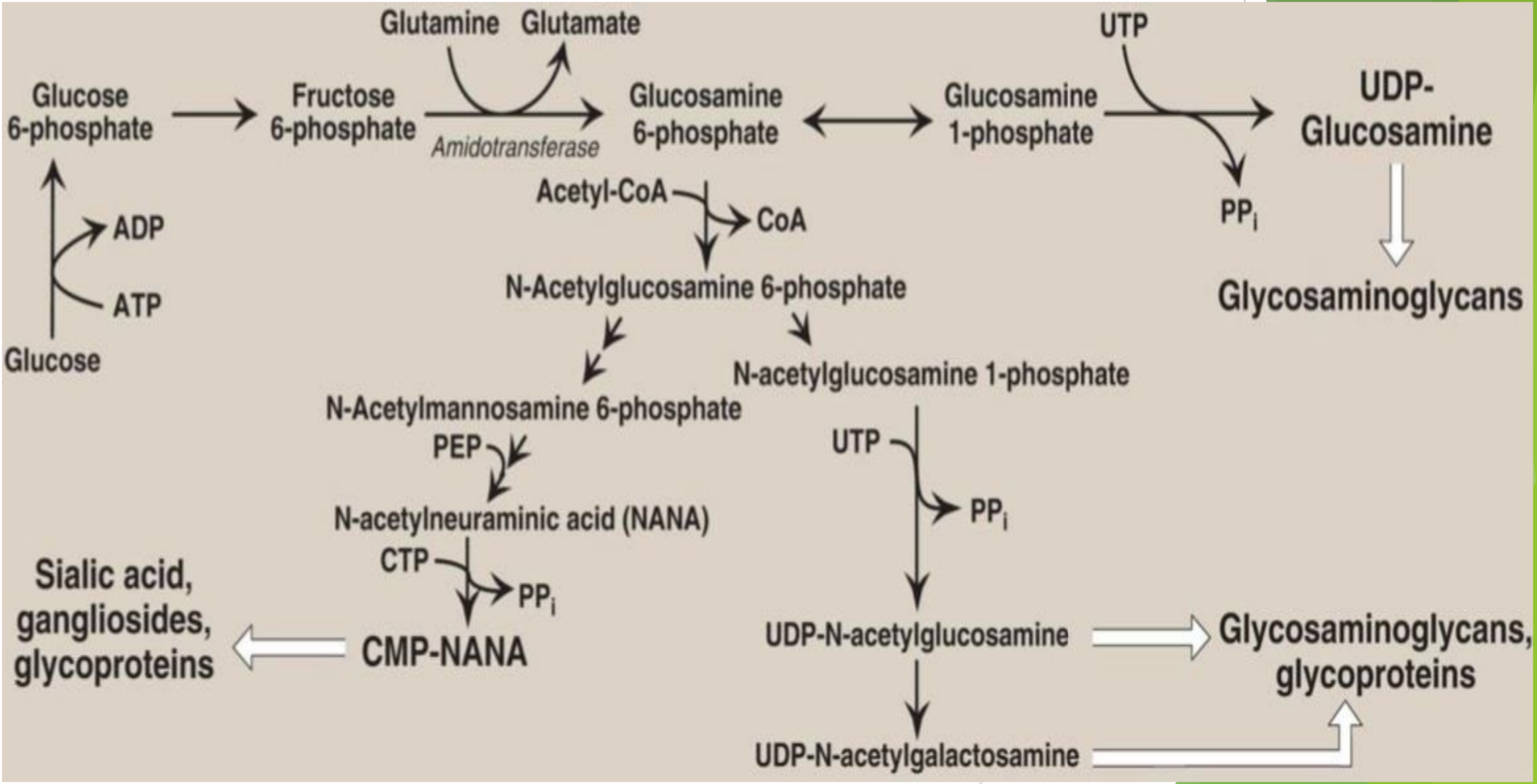
- Sulphated heparin or Heparan sulphate is found in basement membrane and is an essential component of cell surface.



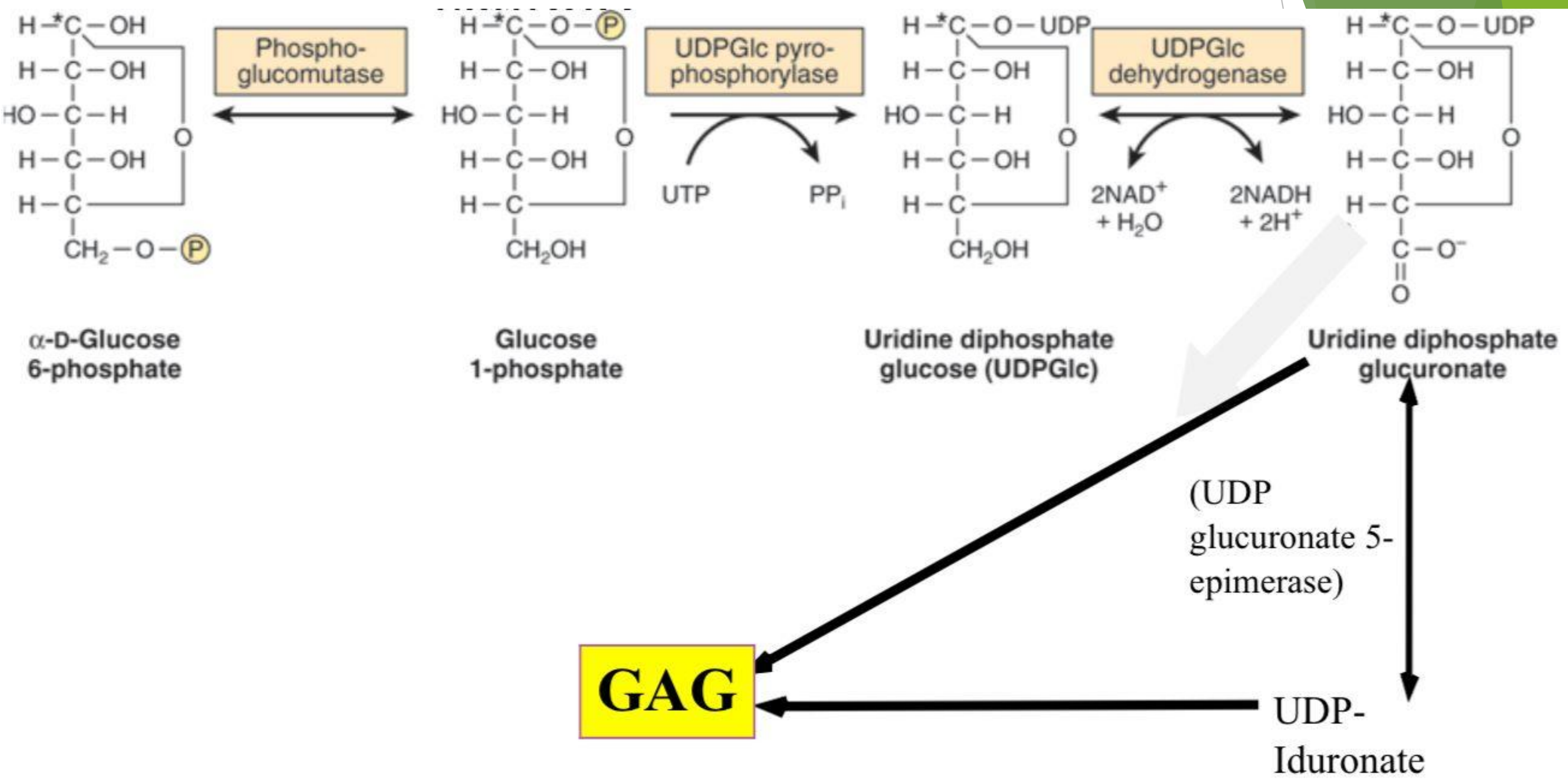
GAG	Sugars	Sulfates	Protein linkage	Location
Hyaluronic acid	GlcNAc, GLcUA	_	None	Skin, synovial fluid, bone, cartilage,vitreous humor, embryonic tissues.
Chondroitin sulfate	GalNAc, GlcUA	GalNAc	Xyl-Ser; associated with HA via link proteins	Cartilage, bone, CNS
Keratan sulfate I and II	GlcNAc, Gal	GlcNAc	GlcNAc-Asn (KS I) GalNAc-Thr (KS II)	Cornea,cartilage, loose connective tissue
Heparin	Gln, IdUA	GlcN GlcN IdUA	Ser	Mast cells, liver, lung, skin
Heparan sulfate	GlcN, GlcUA	GlcN	Xyl-Ser	Skin, kidney basement membrane
Dermatan sulfate	GalNAc, IdUA, (GlcUA)	GalNAc IdUA	Xyl-Ser	Skin, wide distribution

BIOSYNTHESIS OF MUCOPOLYSACCHARIDES

Biosynthesis of Amino sugars



Biosynthesis of Acid sugars



- Biosynthesis of mucopolysaccharides takes place in endoplasmic reticulum and Golgi bodies.

- Biosynthesis of Mucopolysaccharides involves 3 steps :-

  (i) attachment to core proteins

  (ii) chain elongation

  (iii) chain termination

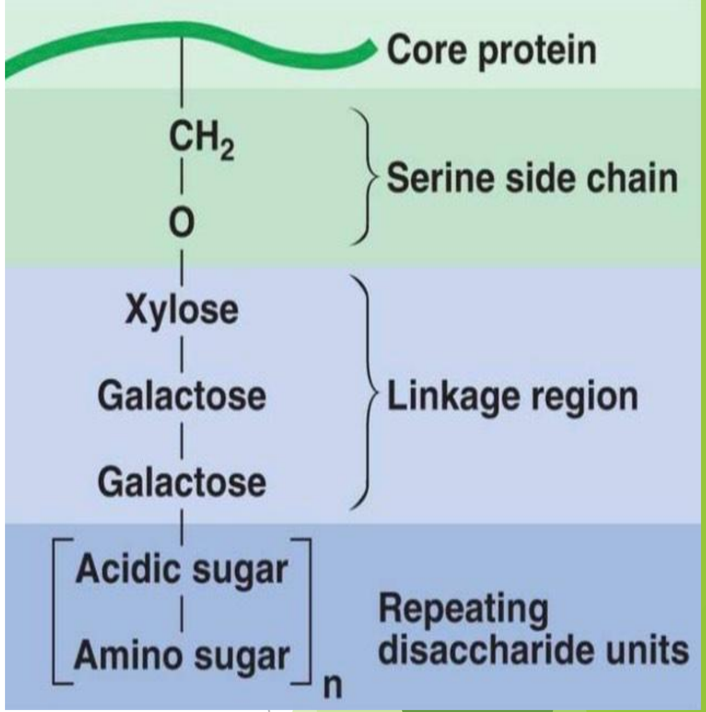
[1] <u>ATTACHMENT TO CORE PROTEINS</u>

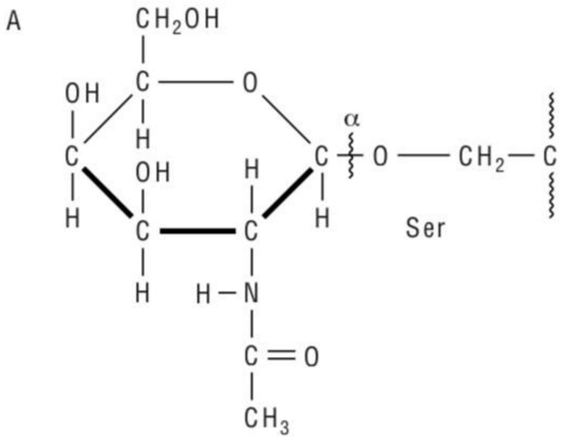
- The linkage between GAGs and their core proteins is generally one of the following three types:-

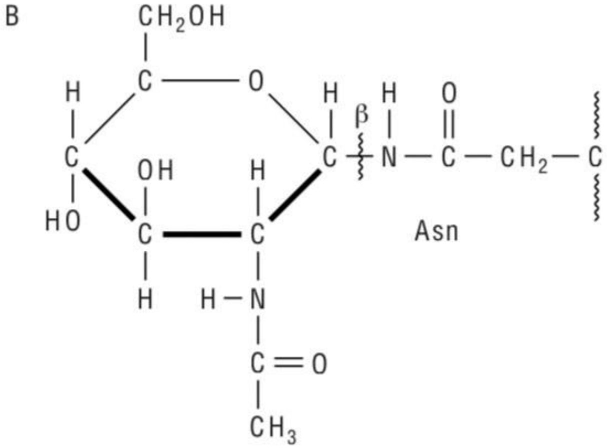
  (i) An O-glycosidic bond between Xylose (xyl) and Ser

  (ii) An O-glycosidic bond between GalNAc (N-acetylgalactosamine) and Ser (Thr)

  (iii) An N-glycosylamine bond between GlcNAc (N-acetylglucosamine) and the amide nitrogen of Asparagine.







[2] CHAIN ELONGATION

- Appropriate nucleotide sugars and highly specific Golgi located Glycosyltransferases are employed to synthesize the oligosaccharide chains of GAGs.
- The "one enzyme, one linkage" relationship appears to hold here.
- The enzyme system involved in chain elongation are capable of high fidelity reproduction of complex GAGs

[3] CHAIN TERMINATION

- This results from

  (1) sulfation, particularly at certain positions of the sugar

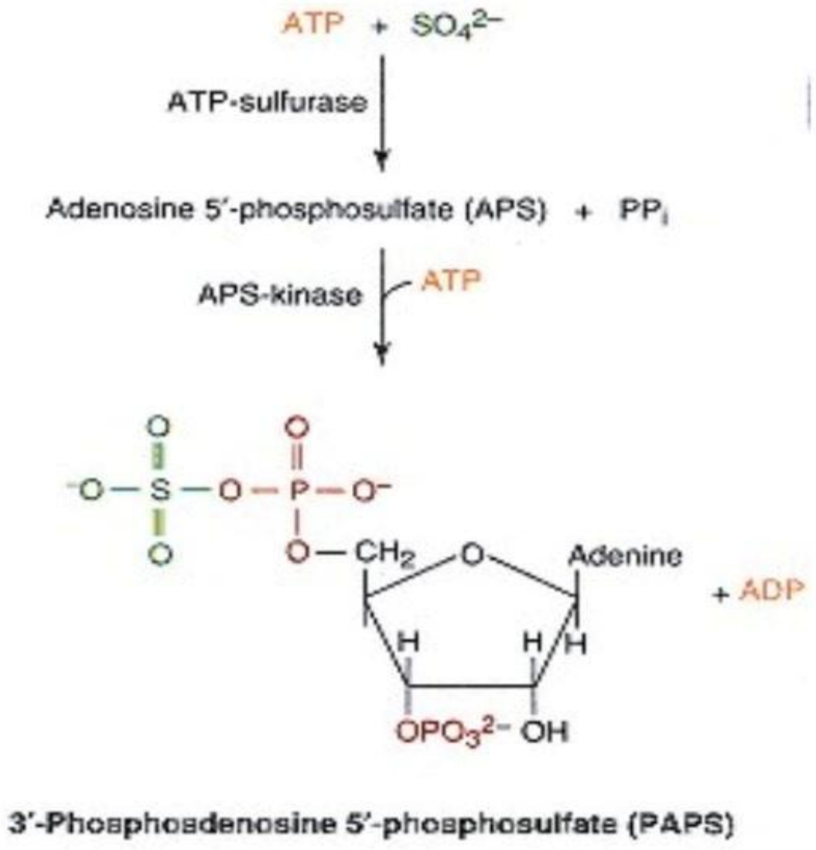
  (2) the progression of the growing GAG chain away from the membrane site where catalysis occurs.

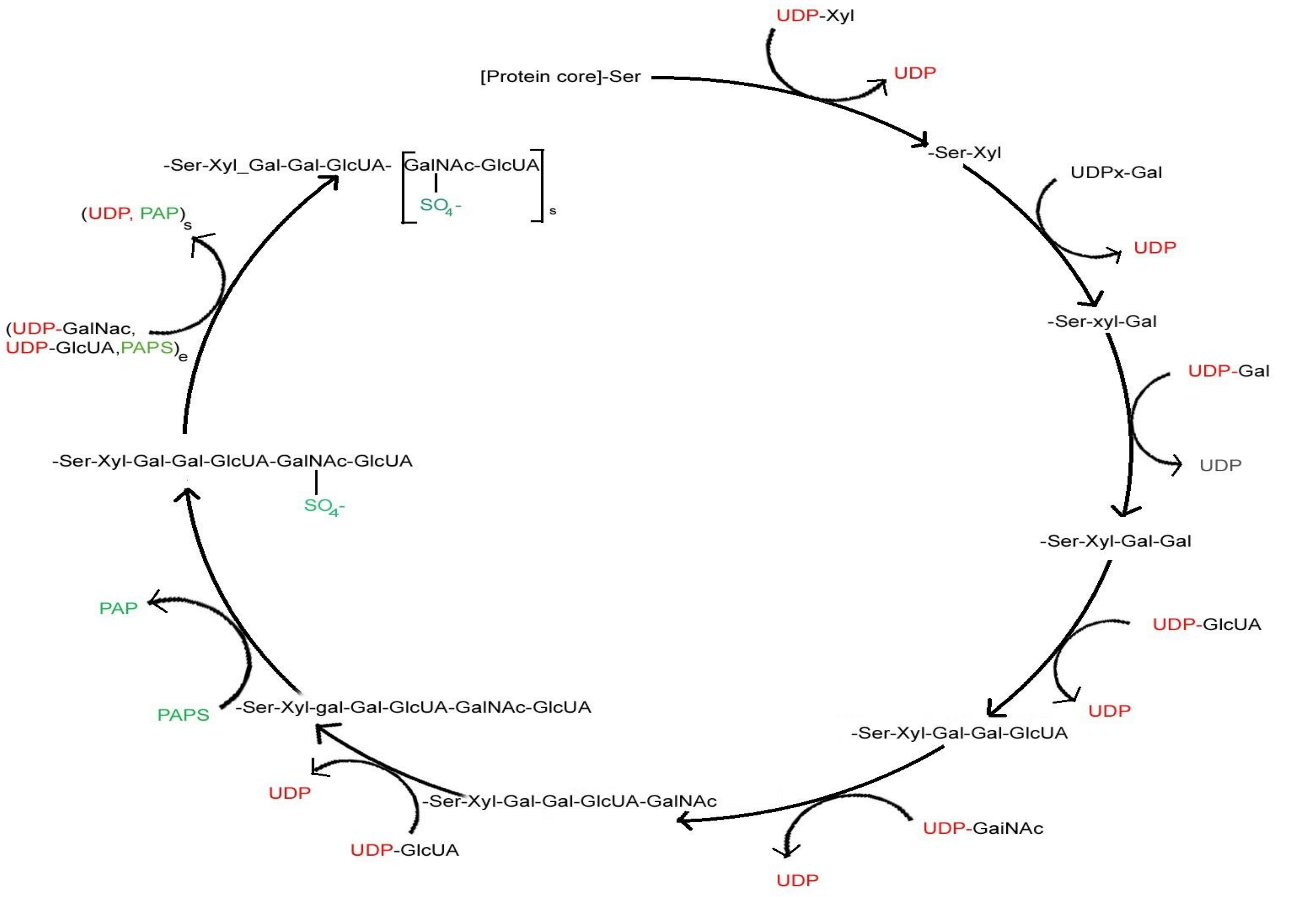
FURTHER MODIFICATIONS

- After formation of the GAG chain, numerous chemical modifications occur :-

  (i) introduction of sulfate groups onto GalNAc and other moieties → catalyzed by sulfotransferases enzyme.

  (ii) the epimerization of GlcUA to IdUA residues → catalyzed by Epimerase enzyme.




[Protein core]-Ser
UDP-Xyl
UDP
-Ser-Xyl
UDPx-Gal
UDP
-Ser-xyl-Gal
UDP-Gal
UDP
-Ser-Xyl-Gal-Gal
UDP-GlcUA
UDP
-Ser-Xyl-Gal-Gal-GlcUA
UDP-GaiNAc
UDP
-Ser-Xyl-Gal-Gal-GlcUA-GalNAc
UDP-GlcUA
UDP
-Ser-Xyl-gal-Gal-GlcUA-GalNAc-GlcUA
PAPS
PAP
-Ser-Xyl-Gal-Gal-GlcUA-GalNAc-GlcUA
SO4-
(UDP-GalNac,
UDP-GlcUA,PAPS)e
(UDP, PAP)s
-Ser-Xyl_Gal-Gal-GlcUA- GalNAc-GlcUA
SO4-
s

FUNCTIONS OF MUCOPOLYSACCHARIDES

(1) GEL-FORMING COMPONENT OF EXTRACELLULAR MATRIX

- GAGs are polyanionic because of presence of numerous sulphate and carboxyl groups on sugar rings.
- Because of polyanionic character, they attract large amounts of water. As a result, they form viscous, hydrated gel that forms a jelly-like matrix (ground substance)
- It fills the extracellular spaces like a packing material.
- It also stabilizes and supports the cellular and fibrous components of tissues.

(2) LUBRICANT ACTIONS

- Due to charge repulsion, the GAG chains tend to slip past one another, in a way two magnets of same polarity do.
- This explains the slippery constituency of mucus secretions and synovial fluid.
- It makes the GAGs suitable for a lubricant action in joints, tendon sheaths and bursae

(3) OTHER FUNCTIONS

- GAGs play an important role in mediating cell-cell interactions.
- Heparin is an inhibitor of blood coagulation.
- GAGs also form a extracellular domains of several membrane proteins.

SUMMARY OF THE FUNCTIONS OF MUCOPOLYSACCHARIDES

- Act as structural components of the ECM
- Have specific interactions with collagen, elastin, fibronectin, laminin, and other proteins such as growth factors
- As polyanions, bind polycations and cations
- Contribute to the characteristic turgor of various tissues
- Act as sieves in the ECM
- Facilitate cell migration (HA)
- Have role in compressibility of cartilage in weight-bearing (HA, CS)
- Play role in corneal transparency (KS I and DS)
- Have structural role in sclera (DS)
- Act as anticoagulant (heparin)
- Are components of plasma membranes, where they may act as receptors and participate in cell adhesion and cell-cell interactions (eg, HS)
- Determine charge-selectiveness of renal glomerulus (HS)
- Are components of synaptic and other vesicles (eg, HS)

MUCOPOLYSACCHARIDOSES

- Like most other biomolecules, GAGs are subject to turnover, being both synthesized and degraded.
- In adults, GAGs generally exhibit relatively slow turnover, their half lives being days to weeks.
- Degradation of GAGs is carried out by lysosomal hydrolases.

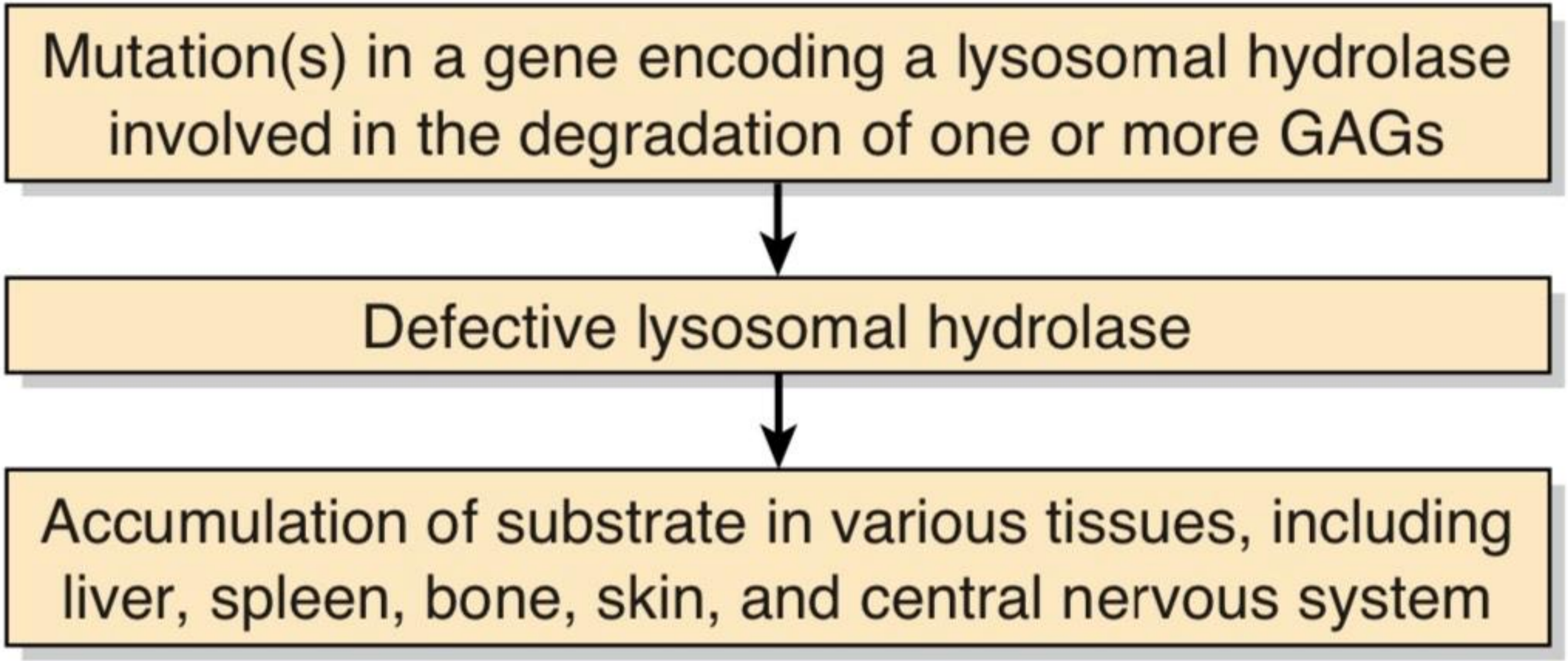
  These include: Endoglycosidases

  Exoglycosidases

  Sulfatases

- These enzymes act in sequence to degrade the various GAGs.

Deficiency of enzymes that degrade GAGs results in Mucopolysaccharidoses.



Major features of Mucopolysaccharidoses

They exhibit a chronic progressive course.
They affect a number of organ systems (ie, they are multisystem disorders).
Many patients exhibit organomegaly (eg, hepato- and splenomegaly may be present).
Patients often exhibit dysostosis multiplex (characterized by severe abnormalities in the development of cartilage and bone, and also mental retardation).
Patients often exhibit abnormal facies (facial appearance).
Other signs sometimes found are abnormalities of hearing, of vision, of the cardiovascular system, and of mental development.

Laboratory tests used in the diagnosis of a Mucopolysaccharidosis

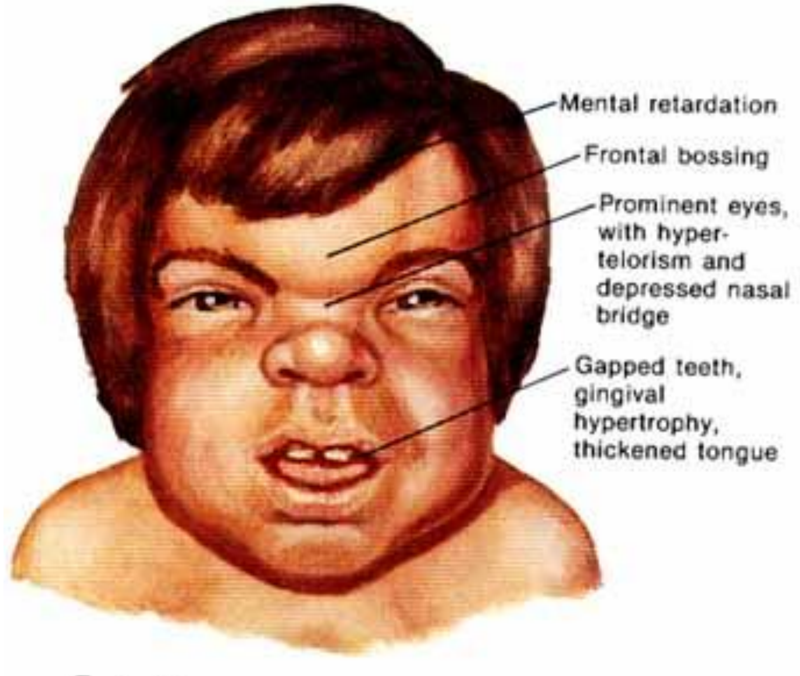
Urinalysis for presence of increased amounts of GAGs.
Assays of suspected enzymes in white blood cells, fibroblasts or possibly serum.
Tissue biopsy with subsequent analysis of GAGs by electrophoresis.
Use of specific gene tests.
Prenatal diagnosis can now be performed in at least certain cases using amniotic fluid cells or chorionic villus biopsy.

HURLER SYNDROME (MPS I)

- It results from an absence of the enzyme alpha-L-Iduronidase.
- Incidence 1 in 100,000 births
- C/F – Facial features → flat face, depressed nasal bridge & bulging forhead

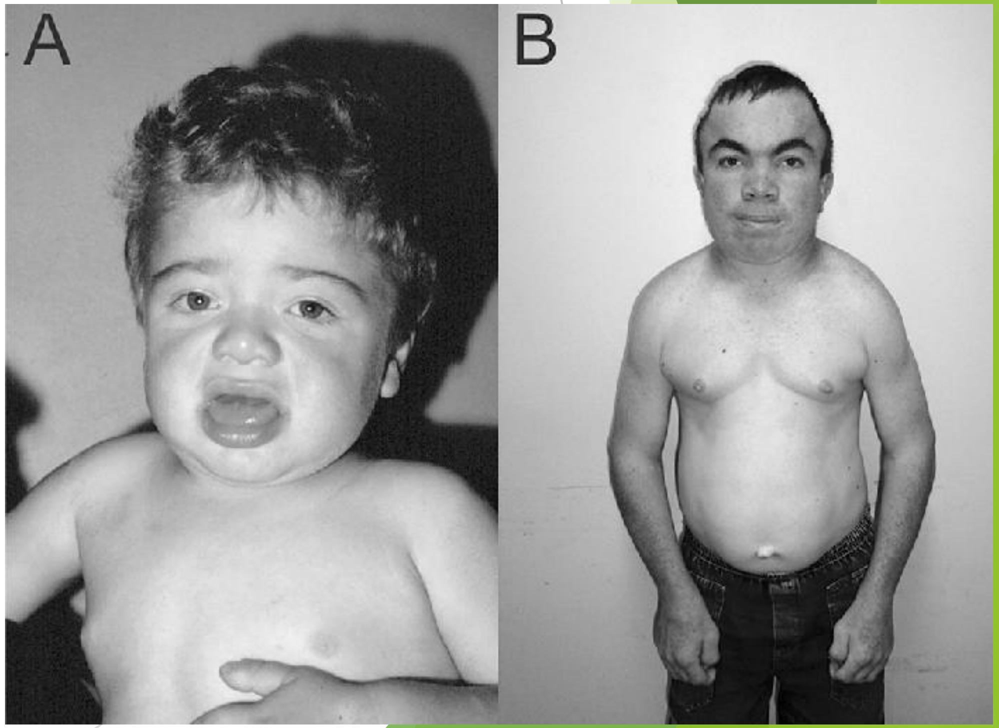
  Cornea → clouded

  Liver, spleen and heart are enlarged
- Child often die before age 10 from respiratory infections and cardiac complications



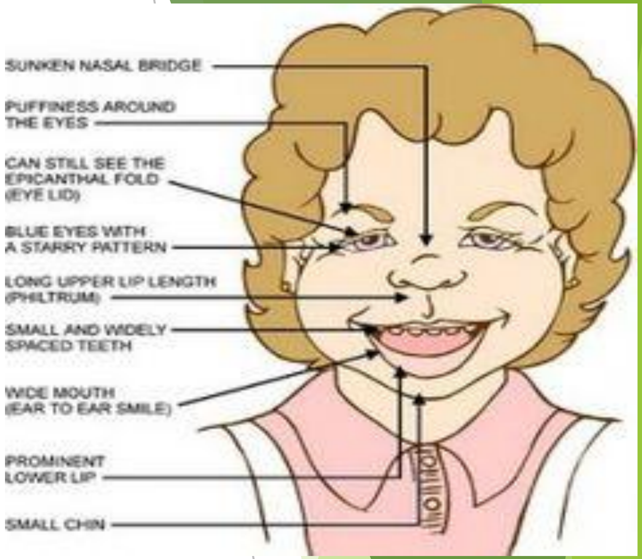
HUNTER SYNDROME [MPS II]

- It is due to deficiency of enzyme iduronate sulfatase.
- X linked recessive inheritance.
- Incidence 1 in 100,000 male births.
- No corneal clouding seen
- Symptoms: Mental retardation, skeletal deformity, deafness seen



SANFILIPPO SYNDROME (MPS III)

- Due to deficiency of enzymes – heparan N-sulfatase
  alpha-N-acetylglucosaminidase
  alpha-glucosaminide acetyltransferase
  N-acetylglucosamine-6-sulfatase



- Incidence : 1 in 70,000 births

- symptoms: progressive dementia, aggressive behaviour, hyperactivity,
  seizures, deafness and loss of vision

MORQUIO SYNDROME (MPS IV)

- It is due to deficiency of enzymes needed to break down the keratan sulfate

  N-acetyl-galactosamine-6-sulfatase

  Beta-galactosidase

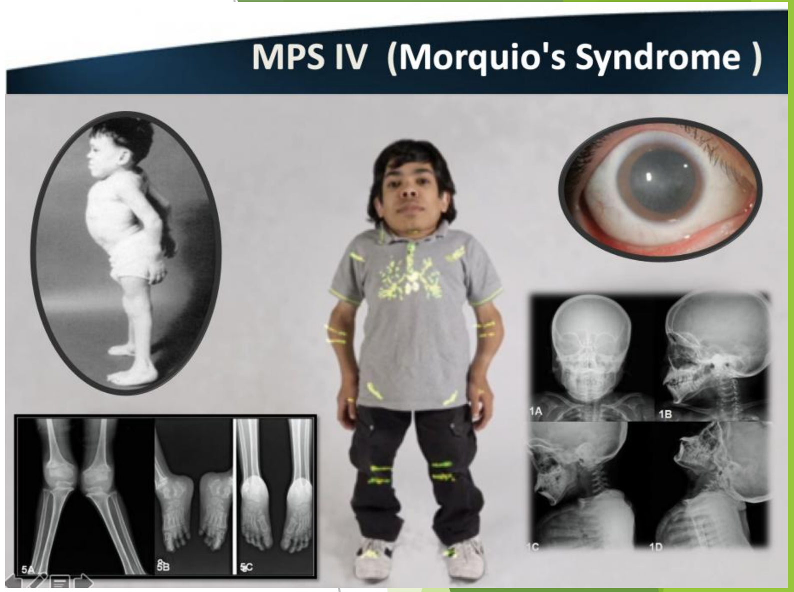
- Incidence – 1 in700,000 births

- Symptoms – spinal root compression resulting from skeletal changes,

  bell shaped chest, shortening of long bones, loss of hearing,

  clouded cornea.

- Intelligence is normal



MAROTEAUX LAMY SYNDROME [MPS VI]

- Due to deficiency of enzyme N-acetylgalactosamine-4-sulfatase.

- Symptoms :- clouded corneas, deafness, skeletal deformities, compressed nerve roots and heart diseases etc.

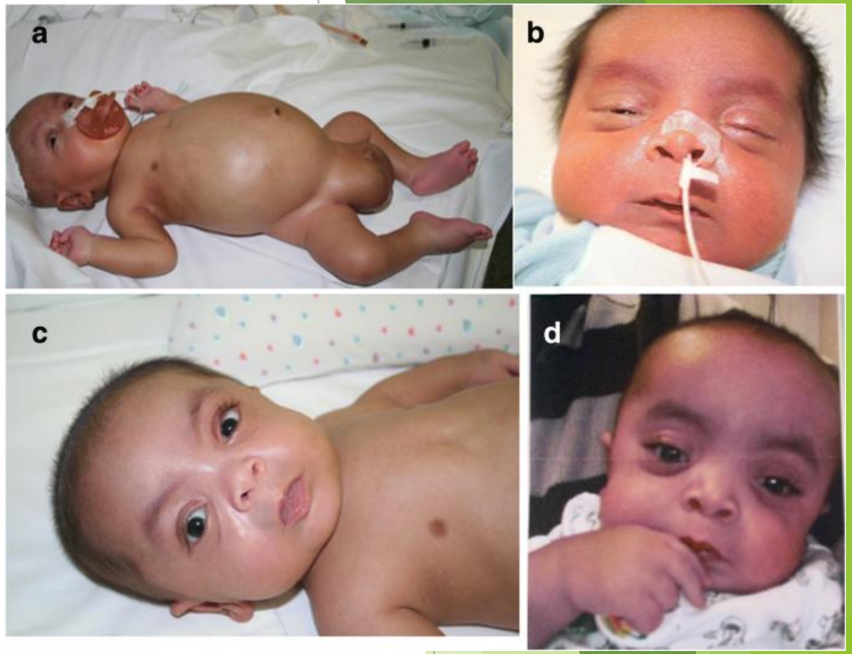
- Children have normal intellectual development.

SLY SYNDROME [MPS VII]

- Caused due to deficiency of the enzyme beta-glucuronidase

- Incidence 1 in 250,000 births

- Symptoms :- Intellectual disability
  corneal clouding
  short stature



NATOWICZ SYNDROME [MPS IX]

- It results from hyaluronidase enzyme deficiency
- Symptoms – nodular soft-tissue masses, bone erosion

SUMMARY OF MUCOPOLYSACCHARIDOSES

Disease name	Abbreviation	Enzyme defect	GAG(s) affected	Symptoms
Hurler, Scheie-hurler-scheie syndrome	MPS I	Alpha-L-Iduronidase	Dermatan sulphate Heparan sulphate	Mental retardation, coarse facial features, hepatosplenomegaly, cloudy cornea
Hunter syndrome	MPS II	Iduronate sulfatase	Dermatan sulphate Heparan sulphate	Mental retardation
Sanfilippo syndrome A	MPS IIIA	Heparan sulfate-N-sulfatase (sulfaminidase)	Heparan sulphate	Delay in development, motor dysfunction
Sanfilippo syndrome B	MPS IIIB	Alpha-N-acetylglucosaminidase	Heparan sulphate	As MPS IIIA
Sanfilippo syndrome C	MPS IIIC	Alpha-Glucosaminide N-acetyltransferase	Heparan sulphate	As MPS IIIA
Sanfilippo Syndrome D	MPS IIID	N-acetylglucosamine 6-sulfatase	Heparan sulphate	As MPS IIIA

Disease name	Abbreviation	Enzyme defects	GAG(s) affected	Symptoms
Morquio syndrome A	MPS IVA	Galactosamine 6-sulfatase	Keratan sulphate Chondroitin sulphate	Skeletal dysplasia, Short stature
Morquio syndrome B	MPS IVB	Beta-Galactosidase	Keratan sulphate	As MPS IVA
Maroteaux-Lamy syndrome	MPS VI	N-Acetylgalactosamine 4-sulfatase (Arylsulfatase B)	Dermatan sulphate	Curvature of the spine, short stature, skeletal dysplasia, cardiac defects
Sly syndrome	MPS VII	Beta-glucuronidase	Dematan sulphate, Heparan sulfate, chondroitin 4-sulfate, chondroitin 6-sulfate	Skeletal dysplasia, short stature, hepatomegaly, cloudy cornea
Natowicz syndrome	MPS IX	Hyaluronidase	Hyaluronic acid	Joint pain, short stature

THANK YOU